TWO FORMULATIONS OF THE STALK COMPARISON OF COHOMOLOGY OVER STRICT HENSELISATIONS AND A SEPARABLY CLOSED FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the cofiltered limit presentation of the base change, the rewrite of the compared fibre product as a limit, and the application of the colimit theorem for cohomology.

Editorial note. The label lemma-check-stalks-better names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's two conditions are displayed formulas of the author, delivered verbatim; the single cited input is kept as an external pointer with its canonical Stacks tag.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Strict henselisations and the identification of the fibre products. Same chapter, the standing notation O^sh_{X,x} and O^sh_{S,s} and the fibre products appearing in both conditions. Supplies the two schemes whose cohomology groups are compared in the statement.
- Cohomology of a cofiltered limit of schemes. Same chapter, the theorem computing the cohomology of the limit as the colimit of the cohomologies along a cofiltered limit with the appropriate hypotheses. Supplies the passage from the limit presentation of the fibre product to the conclusion.

Source reference. Original reference 09YQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7400–7400. Prerequisite pointer invoked at proof line(s) 16578; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\overline{x}$ be a geometric point of X with image $\overline{s}$ in S. Let $\mathrm{Spec}(K) \to \mathrm{Spec}(\mathcal{O}_{S,\overline{s}}^{sh})$ be a morphism with K a separably closed field. Let $\mathcal{F}$ be an abelian sheaf on $\mathrm{Spec}(K)_{\acute{e}tale}$. Let $q \geq 0$. The following are equivalent*

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(1) $H^q(\mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_S \mathrm{Spec}(K), \mathcal{F}) = H^q(\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}}) \times_S \mathrm{Spec}(K), \mathcal{F})$
(2) $H^q(\mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_{\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})} \mathrm{Spec}(K), \mathcal{F}) = H^q(\mathrm{Spec}(K), \mathcal{F})$

Proof. Observe that $\mathrm{Spec}(K) \times_S \mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})$ is the spectrum of a filtered colimit of étale algebras over K. Since K is separably closed, each étale K-algebra is a finite product of copies of K. Thus we can write

$$\mathrm{Spec}(K) \times_S \mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}}) = \lim_{i \in I} \coprod\nolimits_{a \in A_i} \mathrm{Spec}(K)$$

as a cofiltered limit where each term is a disjoint union of copies of $\mathrm{Spec}(K)$ over a finite set A_i. Note that A_i is nonempty as we are given $\mathrm{Spec}(K) \to \mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})$. It follows that

$$\begin{aligned} \mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_S \mathrm{Spec}(K) &= \mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_{\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})} \big(\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}}) \times_S \mathrm{Spec}(K)\big) \\ &= \lim_{i \in I} \coprod\nolimits_{a \in A_i} \mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_{\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})} \mathrm{Spec}(K) \end{aligned}$$

Since taking cohomology in our setting commutes with limits of schemes (Theorem 09YQ) we conclude. □